The two primary ring-model branches are symmetry separated

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Abstract

Veltz and Faugeras numerically conjecture three properties of the two primary pitchfork branches P_1, P_3 in their exact three-dimensional ring model: P_1 lies on the v_3-axis, P_3 lies in the plane $v_3 = 0$, and the branches do not intersect. We prove all three for the primary one-parameter continuations used in the source. Oddness of the centered sigmoid makes the v_3-axis invariant, while reflection symmetry makes the plane invariant. Simple-eigenvalue branch uniqueness places the local pitchforks in those subspaces, and continuation without branch switching preserves them. Finally, the two branches can approach the trivial line only at their distinct bifurcation parameters.

1 Source conjecture

The source paper [1] rewrites the ring model exactly as a three-dimensional system. Let

$$a = \sqrt{J_1}, \qquad c(x) = \cos(\alpha x), \qquad s(x) = \sin(\alpha x), \qquad U_v(x) = v_1 + a v_2 c(x) + a v_3 s(x),$$

on $x \in (-\pi/2, \pi/2)$, with

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} f(x) g(x) \, dx.$$

The source system and its reflection symmetry are shown below.

make the local dynamics sophisticated when the slope parameter λ is big enough.

5.1. Mapping the ring model to the PG-kernel formalism. Expanding the cosine in the previous equation, and denoting by $\cos_\alpha$ (respectively $\sin_\alpha$) the function $x \to \cos(\alpha x)$ (respectively $x \to \sin(\alpha x)$), we find that, depending on the sign of J_i, ($\varepsilon_i = sign(J_i), i = 0, 1$):

$$J = \varepsilon_0 1 \otimes 1 + \varepsilon_1 \sqrt{|J_1|} \cos_\alpha \otimes \sqrt{|J_1|} \cos_\alpha + \varepsilon_1 \sqrt{|J_1|} \sin_\alpha \otimes \sqrt{|J_1|} \sin_\alpha = \sum_{i=0}^{2} \varepsilon_i X_i \otimes X_i,\ \varepsilon_2 = \varepsilon_1.$$

This formulation has the advantage of preserving the symmetries of $\mathbf{W}$. With the notations of the previous section, we have $I^\perp = 0$, and

$$\tau \dot{V}^\parallel = -V^\parallel + W \cdot S(\lambda(V^\parallel + V_0^\perp e^{-t/\tau})) + \varepsilon I^\parallel + \theta$$

where

$$V^\parallel(x,t) = v_1(t) + v_2(t)\sqrt{|J_1|}\cos_\alpha x + v_3(t)\sqrt{|J_1|}\sin_\alpha x, \tag{5.2}$$

i.e. the model is three-dimensional. Note that the previous equation is a rewriting of (5.1) **without any approximation**.

Similarly we have

$$I = I^\parallel = 1 - \beta + \frac{\beta \cos_\alpha x_0}{\sqrt{|J_1|}} \overbrace{\sqrt{|J_1|}\cos_\alpha x}^{X_1} + \frac{\beta \sin_\alpha x_0}{\sqrt{|J_1|}} \overbrace{\sqrt{|J_1|}\sin_\alpha x}^{X_2} \tag{5.3}$$

As $V^\perp(t) \to 0$, we restrict the study to the case $V^\perp = 0$ even if we lose some of the 'real' dynamics by doing so. This is motivated by the fact that the dynamics is made of heteroclinic orbits (as we will see in a moment) between persistent states belonging to the vectpr space $V^\perp = 0$. Hence, using this simplification, we are led to study the following 3D system:

$$\begin{cases} \tau \dot{v}_1 = -v_1 + \varepsilon_0 \langle S(\lambda V), 1\rangle + \varepsilon I_1^\parallel + \theta \\ \tau \dot{v}_2 = -v_2 + \varepsilon_1 \sqrt{|J_1|} \langle S(\lambda V), \cos_\alpha\rangle + \varepsilon I_2^\parallel \\ \tau \dot{v}_3 = -v_3 + \varepsilon_1 \sqrt{|J_1|} \langle S(\lambda V), \sin_\alpha\rangle + \varepsilon I_3^\parallel \end{cases} \tag{5.4}$$

where $\langle f, g\rangle = \int_{-\pi/2}^{\pi/2} f(x) g(x) \frac{dx}{\pi}$, V is given by equation (5.2) and $I_i^\parallel$, $i = 1, 2, 3$ is given by equation (5.3). Note that the basis (X_0, X_1, X_2) is not orthogonal for this inner product.

This system enjoys the symmetries described by the following lemma.

LEMMA 5.1. *When $I_3^\parallel = 0$, if $\mathbf{v} = (v_1\ v_2\ v_3)$ is a solution, then so is $(v_1\ v_2\ -v_3)$. The plane $v_3 = 0$ is invariant by the dynamics.*

Proof. This is a consequence of the fact that $\sin_\alpha$ is an odd function while $\cos_\alpha$ is an even function. □

Figure 1: The exact three-dimensional system (5.4) and source Lemma 5.1, PDF page 20 of [1].

In the simpler case $\mu = \varepsilon = 0$, the nonlinearity is

$$S_0(t) = \frac{1}{1+e^{-t}} - \frac{1}{2} = \frac{1}{2}\tanh(t/2),$$

which is odd, and $v = 0$ is stationary for every slope $\lambda > 0$. For $\varepsilon_0 = -1$, $\varepsilon_1 = 1$, the stationary equation is

$$F(v,\lambda) = \begin{pmatrix} v_1 + \langle S_0(\lambda U_v), 1\rangle \\ v_2 - a\,\langle S_0(\lambda U_v), c\rangle \\ v_3 - a\,\langle S_0(\lambda U_v), s\rangle \end{pmatrix} = 0. \tag{1}$$

For $J_1 > 0$ and α close to but unequal to 2, the source finds two positive simple bifurcation values λ_1, λ_3. The kernel at λ_1 is generated by $(0,0,1)$; the kernel at λ_3 lies in the (v_1, v_2)-plane.

The third eigenvalue σ_2 is negative, and

$$\lambda_i = \frac{4}{\sigma_i}, \qquad \sigma_1 < \sigma_3 \text{ if } \alpha > 2, \qquad \sigma_3 < \sigma_1 \text{ if } \alpha < 2.$$

Thus $\lambda_1 \neq \lambda_3$. The paper then states the numerical conjecture reproduced in Figure 2.

TABLE 5.1

Number of bifurcated Pitchfork branches from $(0, \lambda)$ depending on the values of ε_0, ε_1. The value of α in (5.1) is close to 2.

not orthogonal for this inner product, and hence the matrix $\mathbf{K}$ is not symmetric in this basis. We note σ_1 the eigenvalue of $\mathbf{K}$ corresponding to the eigenvector $(0, 0, 1)$, σ_2 and σ_3 the two eigenvalues ot its upper lefthand 2×2 submatrix. The values, noted λ_i, $i = 1, 2, 3$, corresponding to potential [9] bifurcations are equal to $4/\sigma_i$. The signs of the λ_is give the number of bifurcated branches (recall that $\lambda > 0$). Because $s_2 = S_0^{(2)}(0) = 0$ and $s_3 = S_0^{(3)}(0) = -1/8 \neq 0$, all branches are Pitchfork branches (see section 3.1) whose third order term is $\chi_3^{(i)} = \lambda_i^2 \frac{s_3}{6 s_1} \langle e_i^3, e_i^* \rangle_2 (\approx \lambda_i^2 \frac{s_3}{6 s_1} \|e_i^2\|_2^2 < 0$ for $\alpha \approx 2)$. Hence these branches are directed toward $\lambda > \lambda_i$. This is summarized in table 5.1. The eigenvectors e_i, $i = 1, 2, 3$, of the Jacobian of (5.4) at $\mathbf{v} = 0$ ar given by:

$$e_1 = \sin_\alpha, \quad e_{2,3} = a_{2,3} + b_{2,3} \cos_\alpha$$

We reduce the number of possibilities by assuming from now on that $J_1 > 0$. It turns out that in this case $\sigma_2 < 0$ and there are only two possibilities to consider: $\sigma_1 < \sigma_3$ for $\alpha > 2$ and $\sigma_3 < \sigma_1$ for $\alpha < 2$. This gives the relative position of the different bifurcated branches, noted P_i, $i = 1, 3$. Once we have found the bifurcation point, we can numerically compute the bifurcated branches for all positive values of λ using a continuation method (we used the pseudo-arc length method as described for example in [39, 32]).

From our numerical experiments we conjecture that in the case $\varepsilon_0 = -1$ and $\varepsilon_1 = 1$, P_1 and P_3 satisfy the following properties

1. P_1 lies on the v_3-axis.
2. P_3 lies in the plane of equation $v_3 = 0$.
3. P_1 and P_3 do not intersect.

Remark 11. We can reverse the orientation of the pitchforks by choosing a nonlinearity such that $s_3 > 0$, the bifurcation diagram would be more complex: more saddle node points would appear because of proposition 3.3. It is the fact that $s_2 = 0$ for the sigmoid which produces pichfork branches. Another choice of nonlinearity, for example $S(x) = \frac{1}{1+exp(-x+\epsilon)}$, would produce transcritical branches. It is indeed difficult to imagine that $s_2 = 0$ for an experimental fit of a "real" rate function. But anyway we are not yet looking at the "real" bifurcation diagram since we are assuming $\varepsilon = \mu = 0$.

Figure 5.1 shows a typical example corresponding to the values of the parameters that are found in the largest number of published articles. We come back to this choice in section

Figure 2: The three-item conjecture and its eigenvalue context, PDF page 22 of [1].

The source introduction defines a branch as a one-dimensional set of stationary solutions obtained by varying one parameter. Accordingly, P_i in the quoted passage is the primary pseudo-arclength continuation of the local simple pitchfork at $(0, \lambda_i)$. This is distinct from the larger set obtained by adjoining every possible secondary branch in the same ambient connected component.

2 Proof intuition

The two eigendirections already reveal the correct symmetry spaces. The axis $A = \{(0, 0, t) : t \in \mathbb{R}\}$ represents odd potentials, whereas the plane $E = \{(u, w, 0) : u, w \in \mathbb{R}\}$ represents even potentials. Because S_0 is odd and the integration interval is symmetric, equation (1) restricts to both spaces.

The simple local branch in the full space must therefore coincide with the simple branch obtained by restricting the equation to the relevant invariant space.

This proves the first two geometric claims along every primary continuation that does not switch at a later secondary bifurcation. Since $A \cap E$ contains only $v = 0$, an intersection would have to occur on the trivial line. Linearizing a sequence of nontrivial solutions approaching that line shows that an axis branch can do so only at λ_1, while a positive parameter even branch can do so only at λ_3. Those parameters are distinct.

3 Invariant subspaces and local branches

Lemma 1. *For equation (1), both*

$$A = \{(0,0,t) : t \in \mathbb{R}\}, \qquad E = \{(u,w,0) : u,w \in \mathbb{R}\}$$

are invariant solution subspaces: $F(A,\lambda) \subseteq A$ and $F(E,\lambda) \subseteq E$ for every λ.

Proof. If $v = (0,0,t)$, then $U_v = at\,s$ is odd. Since S_0 is odd, $S_0(\lambda U_v)$ is odd. Its inner products with the even functions 1 and c vanish, so the first two coordinates of F vanish.

If $v = (u,w,0)$, then $U_v = u + aw\,c$ is even. Its composition with S_0 is even, and its inner product with the odd function s vanishes. Thus the third coordinate of F vanishes. The second assertion is also the stationary form of source Lemma 5.1. □

Lemma 2. *Let P_1, P_3 denote the primary one-dimensional continuations, without branch switching, from the simple pitchforks $(0,\lambda_1)$, $(0,\lambda_3)$, respectively. Then*

$$P_1 \subseteq A \times (0,\infty), \qquad P_3 \subseteq E \times (0,\infty).$$

Proof. At $(0,\lambda_1)$, the kernel of D_vF is generated by $(0,0,1)$ and is contained in A. The restriction $F|_A$ has the same one-dimensional kernel and the same transversality condition. The simple-eigenvalue bifurcation theorem [2], applied both to F and to $F|_A$, gives locally unique nontrivial solution curves, up to reparametrization and the two pitchfork arms. The restricted curve is also a full solution curve, so uniqueness identifies it with P_1 locally.

The same argument at $(0,\lambda_3)$ applies to $F|_E$, because the corresponding kernel vector has the form $(a_3, b_3, 0)$.

Continue the two local curves using their restricted equations. At every regular point, the implicit-function uniqueness of the primary arc prevents it from leaving its invariant subspace. At a secondary singular point the restricted arc still gives the distinguished continuation without branch switching. This is exactly the primary pseudo-arclength convention used to name P_1, P_3 in the source passage. Hence the maximal primary continuations remain in A and E, respectively. □

4 Separation of the branches

Lemma 3. *In the positive-λ region, the closure of the nontrivial axis branch can meet the trivial line only at $(0,\lambda_1)$, and the closure of the nontrivial P_3 branch can meet it only at $(0,\lambda_3)$.*

Proof. An axis solution $v = (0,0,t)$ satisfies the scalar equation

$$t = a\,\langle S_0(\lambda a t\,s), s\rangle\,. \tag{2}$$

Suppose $t_n \neq 0$, $t_n \to 0$, and $\lambda_n \to \lambda_* > 0$ along such solutions. Divide (2) by t_n and use $S_0'(0) = 1/4$ and dominated convergence:

$$1 = \lambda_* a^2 S_0'(0) \langle s, s \rangle = \frac{\lambda_*}{4} J_1 \langle s, s \rangle = \frac{\lambda_* \sigma_1}{4}.$$

Therefore $\lambda_* = 4/\sigma_1 = \lambda_1$.

Now let (w_n, λ_n) be nontrivial solutions in E with $w_n \to 0$ and $\lambda_n \to \lambda_* > 0$. After taking a subsequence, $w_n / \|w_n\| \to w \in E$, $\|w\| = 1$. Dividing the equation by $\|w_n\|$ and passing to the limit gives

$$D_v F(0, \lambda_*) w = 0.$$

The restriction of the source matrix to E has eigenvalues σ_2, σ_3, so $\lambda_* \in \{4/\sigma_2, 4/\sigma_3\}$. The source proves $\sigma_2 < 0$; because $\lambda_* > 0$, only $\lambda_* = 4/\sigma_3 = \lambda_3$ is possible. □

Theorem 1. *In the ring-model parameter regime of the source conjecture, the primary branches satisfy:*

$$P_1 \textit{ lies on the } v_3\textit{-axis}, \qquad P_3 \textit{ lies in the plane } v_3 = 0, \qquad P_1 \cap P_3 = \varnothing.$$

Proof. The first two conclusions are Lemma 2. If a point belonged to both branches, its state coordinate would lie in $A \cap E = \{0\}$. By Lemma 3, its parameter would have to equal both λ_1 and λ_3. This is impossible because the source proves that these values are strictly ordered for $\alpha \neq 2$ close to 2. □

5 Verification, scope, and novelty

The proof uses the exact three-dimensional system, not a discretization. Every parity cancellation is on the symmetric interval $(-\pi/2, \pi/2)$. The signs in (1) use precisely $\varepsilon_0 = -1, \varepsilon_1 = 1$. The only limiting argument divides nontrivial solutions before passing to the linearization, so it does not assume that a branch is a graph over λ.

As a non-proof stress test, the script `code/check_roots.py` used 500-point Gauss–Legendre quadrature and 2,000 deterministic random starts at each of 15 slope values from 1 to 200, for the source figure parameters $\alpha = 2.2, J_1 = 1.5$. It found one, then three, then five roots, and no mixed-coordinate root. The command was

```
conda run -no-capture-output -n sandbox python code/check_roots.py.
```

The theorem concerns the named one-dimensional primary continuations, which is the branch convention stated in the source and used by its pseudo-arclength computation. It does not claim that the ambient connected component obtained by adjoining every possible secondary symmetry-breaking branch is contained in the same subspace. That stronger connected-component statement would require excluding secondary bifurcations and is not needed for the literal three-item conjecture.

The four cheap run indexes were searched for arXiv:0910.2247, the exact title, the phrases involving P_1, P_3, the v_3-axis, and ring-model branch intersection. No duplicate was found. Bounded external searches on 2026-08-11 used those exact phrases, the authors, the DOI, and combinations of ring-model symmetry and pitchfork continuation. They found the source and later neural-field literature, but no paper explicitly proving or refuting this three-item conjecture.

Recommended verdict: *candidate full solution, likely valid pending expert review.* The main review point is the match between the source's one-dimensional numerical branch convention and the primary continuation formalized in Lemma 2.

References

[1] Romain Veltz and Olivier Faugeras, *Local/global analysis of the stationary solutions of some neural field equations*, SIAM Journal on Applied Dynamical Systems **9** (2010), no. 3, 954–998; arXiv:0910.2247. https://arxiv.org/abs/0910.2247

[2] Michael G. Crandall and Paul H. Rabinowitz, *Bifurcation from simple eigenvalues*, Journal of Functional Analysis **8** (1971), 321–340.